

C.2

C.2 Quality of Ground Truth Reads

Here, we summarize evidence that the ground truth measure we construct is high quality and robust to various decisions an analyst could make. Recall that the preferred ground truth used throughout the paper is defined using the reads of five board-certified radiologists from Mount Sinai, who each read all 324 patient cases in the study. For each pathology, we aggregate these reports into the ground truth for a patient case i as

$$\omega_i = 1 \left[\sum_{r=1}^5 \frac{\pi_r(\omega_i = 1 | s_{i,r}^E)}{5} > \frac{1}{2} \right]$$

where we suppress the pathology index for simplicity and r indexes the radiologist. This method of aggregating reports is robust to certain types of measurement error and dependence across reports as discussed in [Wallsten and Diederich \(2001\)](#). Table [C.3](#) contains summary statistics for the ground truth created using the Mount Sinai radiologists and a leave-one-out internal ground truth calculated using the reads collected during the experiment under the treatment arm with clinical history but no AI assistance. Table [C.4](#) contains additional summary statistics for the five Mount Sinai ground truth labelers, including their average time and number of clicks. We also show the average agreement of the labels with the original radiologist’s read. Taken together, these analyses demonstrate, for the majority of cases, that the ground truth labelers agree with the assessment of the radiologist who originally read the report in a clinical setting and we can reject that the average probability assessment is equal to 0.5 at the 5% level. Moreover, in Section [C.4.2](#) we show that our results are robust to many different methods of calculating ground truth, including using the experiment leave-one-out ground truth and various aggregation methods of the Mount Sinai reports.

Table C.3: Ground Truth Quality

	Prevalence		Share Rejecting 0.5		Average Number of Rads	
	Sinai	Experiment	Sinai	Experiment	Sinai	Experiment
Top-Level with AI	0.147	0.125	0.696	0.775	5.00	14.04
Pooled with AI	0.043	0.031	0.892	0.932	5.00	14.04
Abnormal	0.194	0.525	0.583	0.568	5.00	14.04
All Pathologies	0.013	0.010	0.953	0.977	5.00	14.04

Note: For each of the pre-registered pathology groups, this table shows the average prevalence, the share of cases where we can reject that $\sum_{r=1}^R \frac{\pi_r(\omega_i=1|s_{i,r}^E)}{R} = 0.5$ at the 5% level, and the average number of reads per case for both the Mount Sinai ground truth labels and the experiment leave-one-out ground truth.

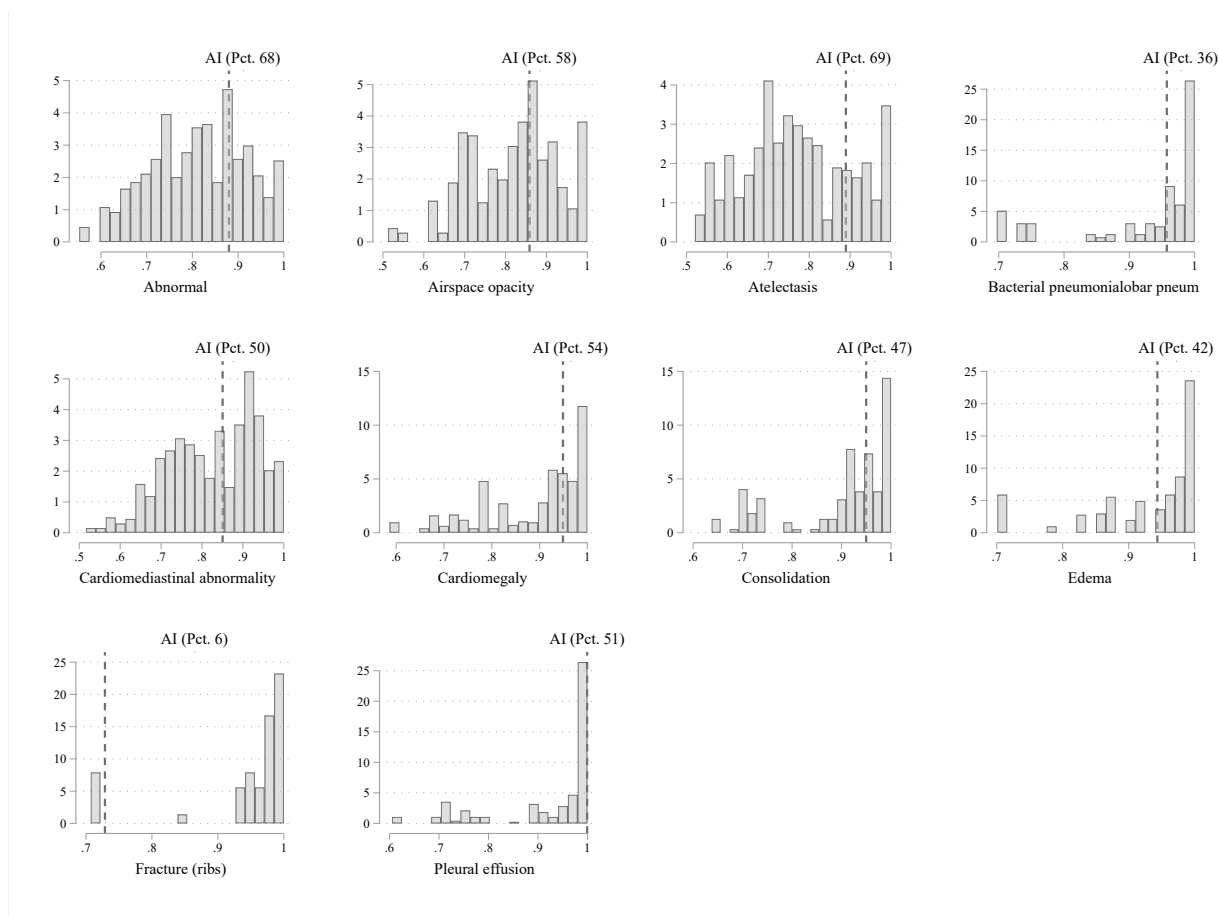
Table C.4: Ground Truth Effort

	Active Time		Clicks		Agreement with Original
	Mean	SD	Mean	SD	
1	77.24	42.78	34.30	17.93	0.868
2	76.44	54.71	32.30	18.24	0.851
3	25.55	30.34	10.84	12.29	0.876
4	79.94	80.22	21.79	20.59	0.866
5	112.96	82.96	26.14	20.12	0.863

Note: For each of the five Mount Sinai radiologists we compute the average and standard deviation of time spent per case and the number of clicks per case. In addition, we compute the average agreement with the original read as labeled by the CheXbert algorithm.

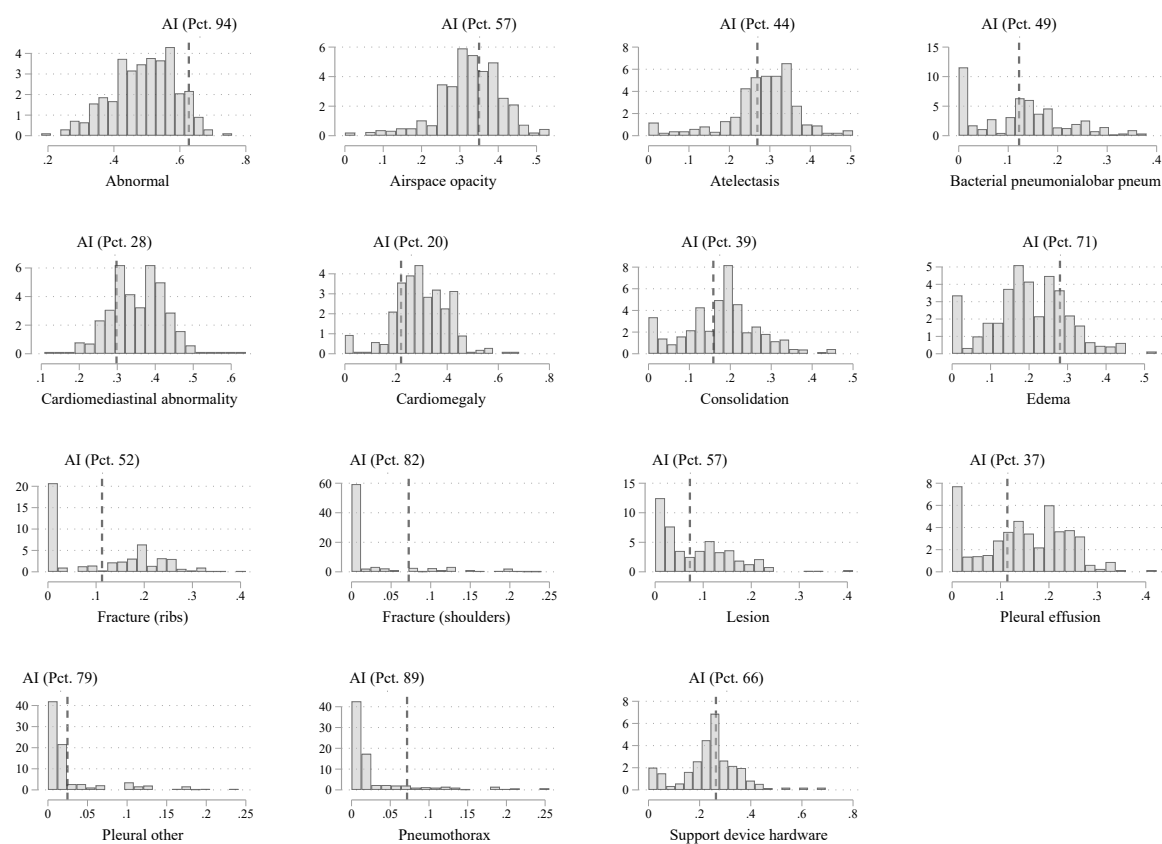
This section presents distributions for two different accuracy measures for radiologists and the AI across different pathology groups. These figures allow for a comparison between the accuracy of the AI relative to the mean radiologist.

Figure C.10: AUROC



Note: This figure summarizes the distribution of radiologist AUROCs across different pathologies, as well as the AUROC of the AI algorithm for the corresponding pathology. Only the cases where contextual history information is available for the radiologist but not the AI prediction were considered.

Figure C.11: RMSE



Note: This figure summarizes the distribution of radiologist RMSE across different pathologies, as well as the RMSE of the AI algorithm for the corresponding pathology. Only the cases where contextual history information is available for the radiologist but not the AI prediction were considered.